

WASP-1b

R-2072

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_121207 · created 2026-08-09T23:19:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456268.6726 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.066 +/- 0.027
Transit depth (Rp/Rs)^2	0.43 +/- 0.36 [%]
Orbital Inclination (inc)	85.11 +/- 1.44
Airmass coefficient 1 (a1)	0.9884 +/- 0.0096
Airmass coefficient 2 (a2)	0.009 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	#1 - [250, 322]
Optimal Aperture	3.59
Optimal Annulus	10.93
Transit Duration (day)	0.132 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.066 ± 0.027 vs 0.1036 (deviation 1.39σ)
Duration fit vs published	0.132 d vs 0.1567 d (deviation 2.23σ)
Mid-time O–C	11.3 min
Depth SNR	2.4
Residual scatter	0.98 %

Light curve

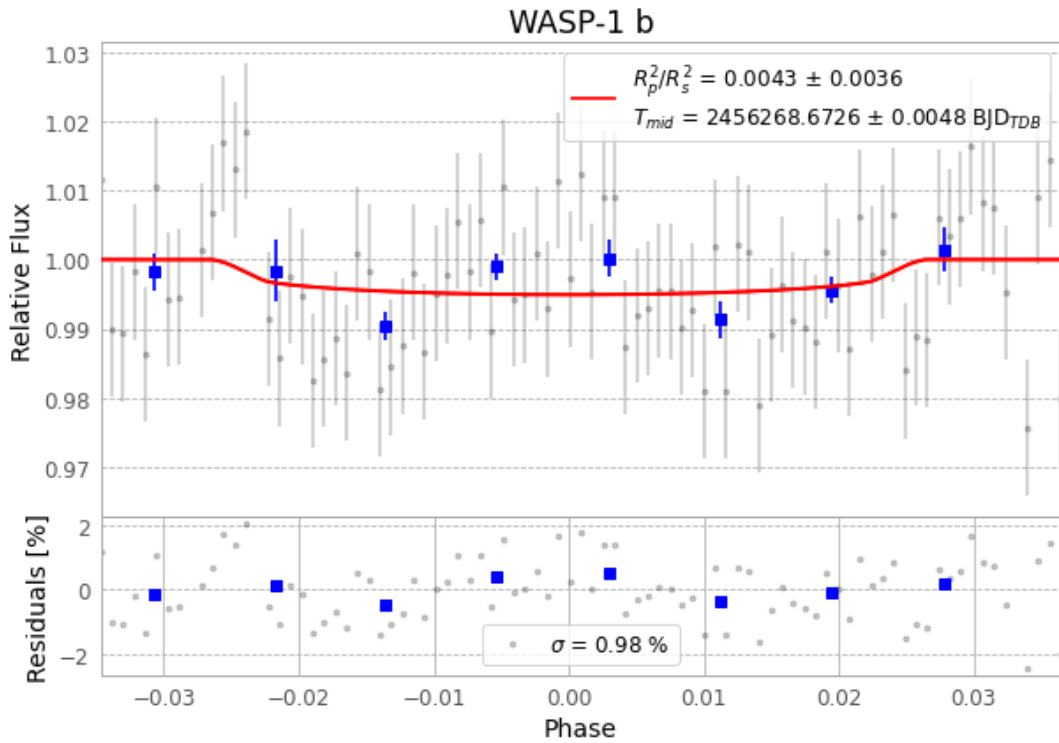


Figure 1 — FinalLightCurve_WASP-1 b_2012-12-06.png (SHA-256 10eed87a16154516...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 38647b60076031d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

87 FITS frames (57 MB), WASP-1121207015713.FITS → WASP-1121207061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-121207.log (23707d2fc754...), FinalLightCurve_WASP-1 b_2012-12-06.png (10eed87a1615...), FinalLightCurve_WASP-1 b_2012-12-06.csv (a288767c090a...)

Compute resources

Wall time 187.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:19Z.